Scalability & Future Plan

Date	30 june 2026
Team ID	SWTID-2026-6540
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses a static CSV dataset and does not retrieve real-time weather or soil data.	Crop recommendations may not reflect current environmental conditions.	high
2	Supports only crop recommendation and does not provide fertilizer or irrigation suggestions.	Users receive limited decision support for complete farm management.	medium
3	No user authentication or prediction history is available.	Users cannot save, review, or track previous crop recommendations.	medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a local Flask server.	Deploy on a cloud platform (e.g., Render or AWS) and use a production web server to handle more concurrent users.

2	Data Storage	Uses a local CSV dataset and a saved ML model (model.pkl).	Integrate a database such as MySQL or PostgreSQL to store user information, prediction history, and future datasets.
3	Performance	Predictions are generated using a single local machine.	Optimize the ML model, implement caching, and upgrade server resources to improve response time and support higher traffic.
4	Security	Basic input validation with no user authentication.	Add user authentication, HTTPS, secure input validation, and role-based access control to improve application security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Requirement analysis, dataset collection, and project planning	Week 1 (Day 1–2)	Establishes a strong foundation for project development.
Phase 2	Data preprocessing, machine learning model training, and model selection	Week 1 (Day 3–4)	Produces an accurate crop recommendation model.
Phase 2	Frontend and backend development using HTML, CSS, Python, and Flask	Week 1 (Day 5–6)	Provides a functional web application for crop prediction.
Phase 3	Model integration, testing, bug fixing, documentation, and final demonstration	Week 1 (Day 7)	Delivers a fully functional and tested OptiCrop application ready for presentation.